Simple Explanation Guide: Synthetic Identity Fraud Detection

Easy-to-Understand Overview of Project Concept, AI Approach, & Results

1. What is this project about? (In Plain English)

Imagine a bank offering instant online loans. A fraudster creates a **'Frankenstein Fake Identity'** using a real PAN tax number combined with fake name, burner phone, and synthetic email. Because the PAN is valid, traditional banks pass the applicant. The fraudster takes the loan money and vanishes!

2. How does OUR AI system catch this fake person?

Our AI acts like a smart detective checking **20 clues across 4 categories**:

1. **KYC Clues:** Name/address mismatch, DOB verification, document image reuse.
2. **Identity Freshness:** Phone line age (<5 days?), email domain age, credit bureau history.
3. **Behavioral Habits:** Humans type normally with backspaces; bots fill form in 5 seconds and copy-paste everything.
4. **Device Telemetry:** Did 5 applications originate from the exact same laptop or Wi-Fi IP address?

3. AI Decision Risk Bands

- ■ **Low Risk (<30%):** Genuine customer → Auto-Approve & Disburse.
- ■ **Medium Risk (30%-60%):** Minor anomaly → Video KYC / OTP Check.
- ■ **High Risk (>60%):** Fake identity → Block Application Immediately!

4. Project Results & Performance

- **Accuracy / ROC-AUC: 91.4%** (High discrimination capability)
- **Fraud Precision: 86.0%** (Low false alarms for real customers)
- **Fraud Recall: 91.1%** (Catches 9 out of 10 fraud attempts)

5. 30-Second Summary for Professor

"Sir, traditional KYC systems only verify static document numbers, so synthetic fake identities bypass them easily. Our project uses Machine Learning on 20 signals—combining document consistency, phone/email age, behavioral biometrics (typing cadence, paste speed), and device velocity. Our model achieves 91.4% ROC-AUC accuracy and is fully deployed with an interactive Streamlit web dashboard for live risk scoring."